Machine Learning

Getting your Kaggle data — Lecture 23 (Chapter 9, Part 4)

Amir Farbin

Part 4 — Getting your Kaggle data

Today: your datasets

- Everyone has picked a competition. We'll open some of them together and read the pages
- Then: getting the data onto your computer, by hand and by command

Reading a competition page

- **Overview:** what is one example? What is predicted? Which **metric** scores you (accuracy, AUC, log-loss)?
- **Data:** which files, how big, what format? Which kind: one simple table, **several tables** linked by IDs, or per-moment data (time series, game-play capture) that must be summarized per example?
- **Rules:** you must **join** (accept the rules) before you can download. Does it still take **late submissions**?
- **Leaderboard:** what do the best get? That's your reference, not your target
- **Code:** other people's public notebooks. Read them to learn; don't copy them

Three ways to get the data

- **Browser:** the Download button on the Data tab. Fine for this project
- **A command in your notebook:** reproducible, so you, or anyone, can get the data again
- **The Kaggle API:** a **REST API** (requests are URLs, replies are data) with a **command-line tool**, `kaggle`, and a Python library
- On **WSL** your browser saves to Windows. From Linux the Windows disk is under `/mnt/c/`:

```
cp /mnt/c/Users/<you>/Downloads/<competition>.zip ~/<your-project>/
```

The Kaggle command-line tool

```
# kaggle 2.2 also needs six on Python 3.13
pip install kaggle six
kaggle auth login --no-launch-browser
```

- `login` prints a link: open it, sign in to Kaggle, paste back the code it shows
- Or make an **API token** at kaggle.com/settings/api, and `export KAGGLE_API_TOKEN=...` or save it in `~/.kaggle/access_token`
- A token is a **password**: never put it in a notebook, a commit or a screenshot

Commands you'll use

```
kaggle competitions list -s mushroom      # search
C=playground-series-s4e8                  # the competition's name
kaggle competitions files $C              # what's there
kaggle competitions download $C -p data --unzip
kaggle competitions submit $C -f submission.csv -m "first try"
kaggle competitions submissions $C        # your scores
```

- The name is the end of the competition's URL: kaggle.com/competitions/**playground-series-s4e8**
- **403 Forbidden** means you haven't accepted the competition's rules: join it on the website first
- A dataset rather than a competition: `kaggle datasets download <owner>/<name> -p data --unzip`

In your notebook

```
!kaggle competitions download playground-series-s4e8 -p data --unzip
!ls -lh data
!head -3 data/train.csv
import pandas as pd
df = pd.read_csv("data/train.csv")
```

- Look before you load: the header row, the separator, the number of rows (`wc -l`), the size on disk
- **Keep the data out of Git**: add `data/` to `.gitignore`. Your README says where the data comes from, and the command gets it

The submission

- `test.csv` has the inputs and an **ID** column, but no answers. Keep the IDs: the submission needs them
- Predict every row of `test.csv`, write it in `sample_submission.csv`'s format, and submit
- Kaggle scores it against answers only it has. Late submissions get a score but no rank
- Before predicting, retrain your chosen model on all of `train.csv`

Vocabulary

- **competition, leaderboard, metric, rules, late submission, public notebook**
- **REST API, command-line interface (CLI), API token, OAuth login**
- **WSL, `/mnt/c/`, `.gitignore`, reproducible download**
- **`train.csv`, `test.csv`, `sample_submission.csv`, ID column, submission**